

THE INTERNATIONAL MUSTACHE AUTHORITY

★ Facial Hair Research Division ★

TensorFlow • CNN Neural Networks

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I.M.A.
INTEGRITY.
METRICS.
MUSTACHES.





CELEBA DATASET

CELEBFACES ATTRIBUTES DATASET



202,599
IMAGES



40
FACIAL ATTRIBUTES

DATASET STATUS
VERIFIED



INTERNATIONAL MUSTACHE AUTHORITY



RESEARCH DIVISION



ATTRIBUTE CATEGORIES



DISTRIBUTION OVERVIEW



SOURCE

CELEBA
v1.0





MODEL 1: MUSTACHE DETECTION

FIRST CLASSIFIER: MUSTACHE OR CLEAN

DATASET CREATION

SOURCE: CELEBA (MALE ONLY)

```
mustache_df = men[men['Mustache'] == 1]
clean_df     = men[men['Mustache'] == -1]
```

```
mustache_sample = mustache_df.sample(
    n=4000, random_state=42)
clean_sample     = clean_df.sample(
    n=4000, random_state=42)
```

Total män: {len(men);,}

MUSTACHE: 4,000

CLEAN: 4,000

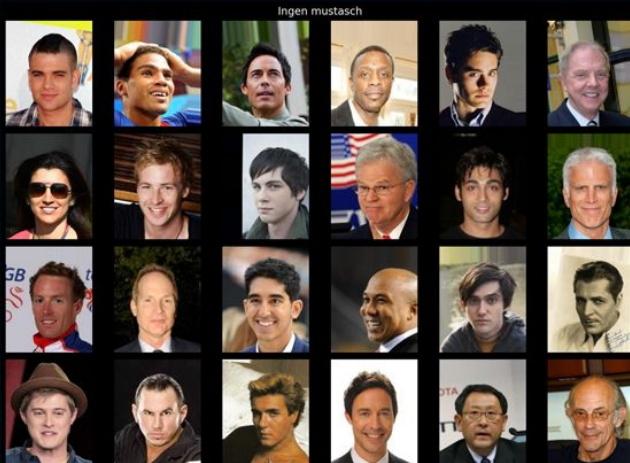
DATA LOADING (TENSORFLOW)

```
train_ds = tf.keras.utils.image_dataset_from_directory(
    'data/dataset_v2',
    validation_split=0.2,
    subset='training',
    seed=42,
    image_size=(128, 128),
    batch_size=32,
    label_mode='binary')
```

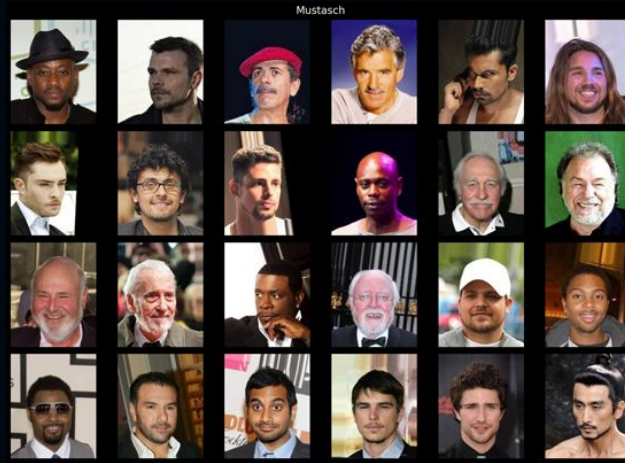
```
val_ds = tf.keras.utils.image_dataset_from_directory(
    'data/dataset_v2',
    validation_split=0.2,
    subset='validation',
    seed=42,
    image_size=(128, 128),
    batch_size=32,
    label_mode='binary')
```

Klasser: ['clean', 'mustache']

CLEAN (INGEN MUSTASCH)



MUSTACHE



8,000
TOTAL IMAGES



4,000
CLEAN



4,000
MUSTACHE



80%
TRAINING
20%
VALIDATION



BINARY
CLASSIFICATION
OUTPUT: 2 CLASSES



EXPERIMENT STATUS: INITIALIZED



MODEL 1: EVALUATION

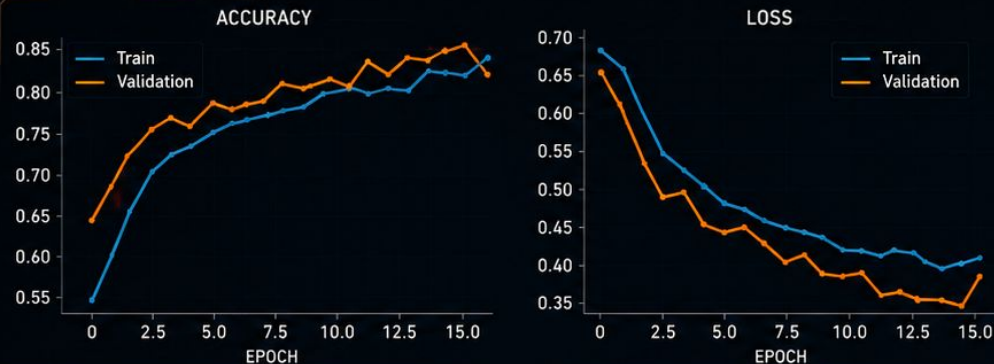
MUSTACHE DETECTOR CNN



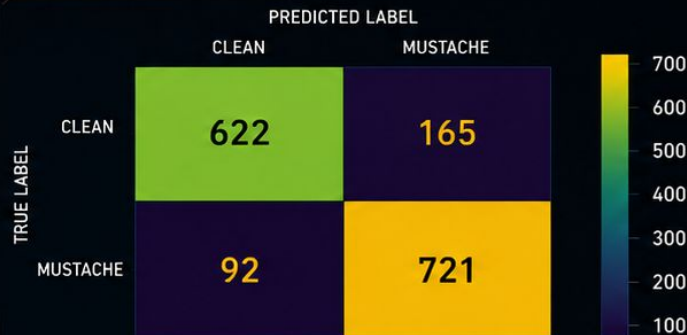
IMA-17A

CLEARANCE LEVEL: GOLD

TRAINING HISTORY



CONFUSION MATRIX



83.9%

VALIDATION ACCURACY



15

EPOCHS



BINARY CLASSIFICATION

OUTPUT: 2 CLASSES



622

CORRECT
CLEAN PREDICTIONS

721

CORRECT
MUSTACHE PREDICTIONS

165

FALSE
POSITIVES

92

FALSE
NEGATIVES

TRAINING CONFIGURATION

```
model_v2.compile(  
    optimizer='adam',  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)  
early_stop = tf.keras.callbacks.EarlyStopping(  
    monitor='val_loss',  
    patience=5,  
    restore_best_weights=True,  
    verbose=1  
)
```

```
history = model_v2.fit(  
    train_ds,  
    validation_data=val_ds,  
    epochs=15,  
    callbacks=[early_stop]  
)
```

IMA-17A

CLEARANCE LEVEL: GOLD

★ SCIENCE. DISCIPLINE. DIGNITY. MUSTACHES. ★



RESEARCH LOG

DATE: 2024-05-22

AUTH: IMA-7862-6
STATUS: APPROVED



★ RESEARCH DIVISION ★

★ CLASSIFIED RESEARCH MATERIAL ★

MANUAL EPICNESS CLASSIFICATION

The Burden of Science

EVERY MUSTACHE
TELLS A STORY.
ONLY THE EPIC ONES
MAKE HISTORY.



4,414

MUSTACHES WERE EXAMINED

597

WERE DEEMED WORTHY



NEW EPIC DATASET

EPIC
334

THIN
263

TOTAL IMAGES
597

EPICNESS EVALUATION FORM

- ☒ THICKNESS
- ☒ DENSITY
- ☒ SHAPE
- ☒ SYMMETRY
- ☒ OVERALL IMPACT



REJECTED
4,414

MUSTACHE
REPORTS

FUELING
GREAT
SCIENCE

INTERNATIONAL
MUSTACHE AUTHORITY

IMA-17A

CLEARANCE LEVEL: GOLD

★ EVERY IMAGE WAS MANUALLY REVIEWED BY THE INTERNATIONAL MUSTACHE AUTHORITY ★

RESEARCH LOG

DATE: 2024-05-22



• RESEARCH DIVISION •

• CLASSIFIED RESEARCH MATERIAL

MODEL 2: EPIC MUSTACHE CLASSIFIER

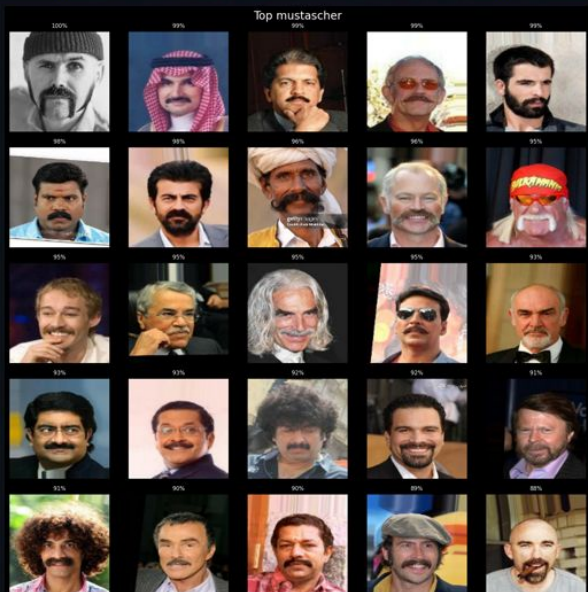
EPIC DETECTOR — MOBILENETV2



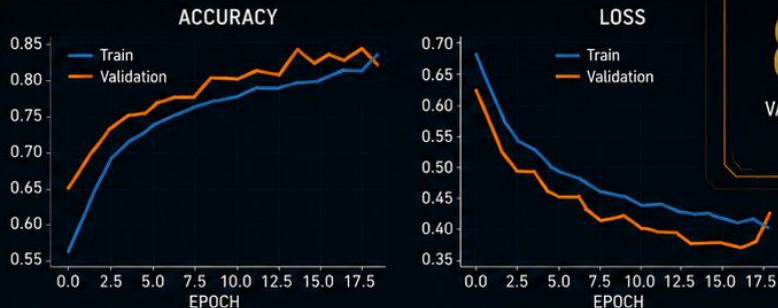
IMA-17A

CLEARANCE LEVEL: GOLD

TOP 25 MOST EPIC MUSTACHES



TRAINING HISTORY



VALIDATION ACCURACY

86.6%

VALIDATION LOSS: 0.3694



CONFUSION MATRIX (VALIDATION SET)

		PREDICTED LABEL	
		EPIC	THIN
TRUE LABEL	EPIC	57	9
	THIN	7	46

NEW EPIC DATASET

 EPIC
334 THIN
263

TOTAL IMAGES

597BUILT ON MANUALLY CURATED DATA
EVERY IMAGE REVIEWED BY EXPERTSBINARY CLASSIFICATION
EPIC = 1 | THIN = 0MISSION: IDENTIFY TRUE EPICNESS
ONE MUSTACHE AT A TIME

SCIENCE. DISCIPLINE. DIGNITY. MUSTACHES.



• RESEARCH DIVISION •

• CLASSIFIED RESEARCH MATERIAL

MODEL 2: EPIC MUSTACHE CLASSIFIER

EPIC DETECTOR — MOBILENETV2



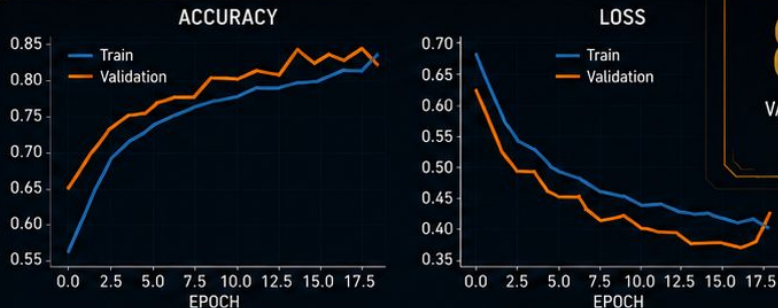
IMA-17A

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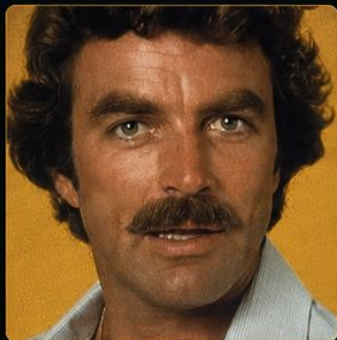
SCIENCE. DISCIPLINE. DIGNITY. MUSTACHES.



SUBMITTED SPECIMEN

Official Facial Hair Assessment by the International Mustache Authority

Submit specimen for analysis



Submitted specimen

Initiate Analysis

ANALYSIS RESULTS

mustache_prob: 0.9140045046806335

thin_prob: 0.0927349105477333

epic_score: 0.9979915983974934

MUSTACHE POWER RATING™

99.8 / 100



LEGENDARY STACHE

The International Mustache Authority is impressed.

Results certified by the International Mustache Authority™



IMA-17A

CLEARANCE LEVEL: GOLD



UPLOAD



ANALYZE



RESULTS



REPORT

RESEARCH LOG

DATE: 2024-05-22

SCIENCE.
DISCIPLINE.
DIGNITY.
MUSTACHES.

IMA-17A

EVERY MUSTACHE
TELLS A STORY.
ONLY THE EPIC ONES
MAKE HISTORY.INTERNATIONAL
MUSTACHE AUTHORITY

TOP SECRET

SPECIMEN ANALYSIS REPORT

EPIC